Response to Reviewers

Texture Resonance Retrieval for Retrieval-Grounded Editable Audio Effect Preset Selection

IEEE Transactions on Multimedia — Major Revision

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We sincerely thank the reviewers for their detailed and constructive feedback. Below we address each concern point by point, organized by the major (M1–M5) and minor (m1–m8) issues identified in the review. All referenced section, table, and figure numbers refer to the revised manuscript.

Major Issues

M1: Limited Dataset Scale and Near-Duplicate Risk

Reviewer Comment: The benchmark comprises only 1,267 guitar presets (204 test queries), which is small by retrieval standards. After removing exact audio reuse, 200 near-duplicate pairs remain. This raises the concern that strong retrieval performance may be inflated by trivially close knowledge-base items.

Our Response: We agree that dataset scale is a genuine limitation and have taken concrete steps to quantify its impact on the reported results:

1. **Near-duplicate sensitivity analysis** (new Protocol-A paragraph "Near-duplicate sensitivity"): We progressively remove near-duplicates from the knowledge base at four L2 thresholds ($\tau \in \{0.005, 0.01, 0.02, 0.05\}$) and re-evaluate TRR. The normalized L2 degrades as the KB shrinks from 1,063 to 83 candidates, so the revised paper treats database density as a real boundary condition rather than as a solved issue. Full details appear in the supplementary sensitivity subsection.

2. **Near-duplicate detection infrastructure**: We provide a reusable `near_dup_filter.py` script and `near_dup_sensitivity.py` experiment, enabling reviewers to reproduce the analysis at arbitrary thresholds.

3. **Explicit scope acknowledgment**: We clearly state in the experimental setup, Discussion, and Limitations that this is a pilot benchmark on guitar effects, not a claim of universal scale.

Change in manuscript: Protocol-A near-duplicate sensitivity paragraph and table added; supplementary sensitivity analysis retained; Abstract revised to mention near-duplicate sensitivity.

M2: Weak Baseline Coverage

Reviewer Comment: The paper compares TRR only against text-RAG, FeatureNN-RAG, Wav2Vec-RAG, and CLAP. Strong audio representation baselines such as PANNs and PaSST, and a direct regression baseline (MLP), are absent.

Our Response: We have added three new baselines:

1. **PANNs (CNN14)**: Large-scale pretrained audio pattern recognition network [Kong et al., 2020]. We use the CNN14 variant with AudioSet-pretrained embeddings for retrieval.

2. **PaSST**: Efficient audio transformer with Patchout [Koutini et al., 2022]. Retrieval uses the AudioSet-pretrained embedding space.

3. **MLP-Regressor**: A two-layer MLP with 256 hidden units trained on mean-pooled Wav2Vec2 features to directly predict flattened parameter vectors, establishing a supervised regression boundary.

All retrieval baselines are evaluated under Protocol-A with the same split and metrics. Results are reported in Table 3 of the main paper, while the MLP boundary is reported separately because it predicts dense parameter leaves rather than retrieving executable presets.

Change in manuscript: Table 3 expanded with PANNs and PaSST retrieval rows; direct-regression boundary table added; baseline descriptions updated.

M3: No Ablation Studies

Reviewer Comment: There are no ablation studies validating the core design choices of TRR: projection dimensionality, layer selection, and projection type (PCA vs. random).

Our Response: We have added a controlled supplementary mechanism-ablation subsection with three diagnostic comparisons:

1. **Projection dimensionality and layer selection**: Supplementary Table S2 reports controlled mechanism rows including d=32 and d=64 projected multi-layer variants, layer 5 Gram aggregation, and the full unprojected Gram boundary.

2. **Aggregation and normalization**: Supplementary Table S2 compares same-backbone mean pooling, Gram aggregation, and a no-normalization variant. This isolates the contribution of second-order aggregation without claiming global architectural optimality.

3. **Interpretation guardrail**: The main paper now cites the ablation only as mechanism support for TRR as a retrieval prior, not as proof that the chosen layer set and projection dimension are globally optimal.

Change in manuscript: Supplementary mechanism-ablation subsection and Table S2 added; Algorithm 1 updated to specify "frozen random linear projection" and correct dimension to 32; Abstract mentions mechanism ablations.

M4: Unnormalized Metrics

Reviewer Comment: All L2 error metrics operate on raw parameter values with heterogeneous physical scales. A compressor threshold in dB and a delay time in ms have very different magnitudes, making raw L2 hard to interpret and potentially dominated by large-scale parameters.

Our Response: We now report a **Normalized L2** (Norm. L2) metric alongside the original raw L2:

1. Each parameter leaf is min-max normalized to $[0, 1]$ using the physical DSP parameter ranges defined in the plugin source code (`PluginAudioParameters.h`).

2. The ranges are documented in a shareable `param_ranges.json` file for reproducibility.

3. The evaluation pipeline (`evaluate.py`) supports both modes: `Evaluator(normalize=False)` for backward compatibility and `Evaluator(normalize=True)` for the new normalized metrics.

Change in manuscript: Section 4.2 "Normalized L2" paragraph (new); Table 3 adds a Norm. L2 column; `param_ranges.json` and updated `evaluate.py` provided.

M5: Listening Test Methodological Weaknesses

Reviewer Comment: Trials 2–5 compare the system against a "manual" baseline with unknown provenance and time budget. There are also no demographic details about participant expertise.

Our Response: We have addressed this in several ways:

1. **Condensed Trials 2–5**: The main text now reports aggregate trial-block summaries with explicit caveats about manual-baseline provenance, rather than treating the manual comparison as a fairness-critical optimization benchmark.

2. **Participant metadata boundary**: We report the available age and gender metadata and explicitly state that the logs do not preserve device metadata, expertise, or manual-baseline time budget.

3. **Local vLLM diagnostic**: Following the revision request to add evidence where feasible, we added a conservative local qwen3.5-9b vLLM diagnostic that compares text-side perceptual-recognizability scores with Trial-1 human ratings. The result is negative and is reported as a boundary condition rather than as supportive perceptual evidence.

4. **Interpretation guardrails**: We explicitly frame the listening test as exploratory perceptual context, not as validation of the objective metrics or proof of superiority over MusicGen.

Change in manuscript: Section 4.5 condensed and reframed; local vLLM diagnostic added; missing participant-expertise and manual-baseline provenance now stated as limitations.

Minor Issues

m1: Unclear Algorithm 1 Projection Description

Reviewer Comment: Algorithm 1 describes a "linear projection $P : \mathbb{R}^{768} \to \mathbb{R}^{64}$" but the code uses d=32.

Our Response: Fixed. Algorithm 1 now correctly states "frozen random linear projection $P : \mathbb{R}^{768} \to \mathbb{R}^{32}$" with Xavier initialization and no gradient updates. All downstream dimension references (Gram matrix 32×32, embedding 1024-dimensional) are consistent.

Change in manuscript: Algorithm 1 corrected; prose in Section 3.2 updated.

m2: Over-Hedging

Reviewer Comment: Phrases like "currently evaluated," "we therefore interpret," and "on the current benchmark" appear excessively, reducing readability.

Our Response: We conducted a systematic pass to reduce redundant qualifiers while preserving scientific rigor. Key reductions include removing unnecessary scope modifiers, simplifying "We therefore interpret X as..." constructions, and keeping only the qualifiers needed to state the benchmark boundary honestly.

Change in manuscript: Throughout the manuscript; redundant hedging reduced while retaining necessary scope boundaries.

m3: Section 3.3 (Preliminary Personalization Extension) is Unreferenced

Reviewer Comment: Section 3.3 describes a personalization system but no results support it, and it contributes to page bloat.

Our Response: Deleted entirely. The Discussion now explicitly excludes personalization from validated claims.

Change in manuscript: Section 3.3 removed.

m4: Table 1 (Parameter–Waveform Duality) Lacks Data

Reviewer Comment: Table 1 contains only a conceptual comparison with no empirical data and consumes valuable page budget.

Our Response: Table 1 has been removed. The Parameter–Waveform Duality concept is retained as a brief inline paragraph in Section 2.

Change in manuscript: Table 1 removed; inline description preserved.

m5: Missing Visualization

Reviewer Comment: No visualizations of the TRR embedding space or Gram matrix structure are provided.

Our Response: We have added two figures:

1. **t-SNE diagnostic**: comparison of TRR vs. Wav2Vec2 mean-pooled embeddings, colored by style label.

2. **Gram-matrix diagnostic**: representative Gram-matrix examples showing the second-order correlation structure captured by TRR.

Change in manuscript: Two vector diagnostic figures added and referenced in Section 3.2.

m6: BibTeX Errors

Reviewer Comment: Several BibTeX entries have wrong types (e.g., `@inproceedings` for journal articles like Neuron), incomplete author lists ("and others"), and missing metadata.

Our Response: Targeted BibTeX cleanup performed for the entries most likely to affect reviewer trust: `LLM2Fx` is no longer represented as an arXiv "proceedings" item, its author list and arXiv DOI are expanded, `AudioLDM` is represented as an ICML 2023 paper, and `MusicLM`'s arXiv entry now uses the full arXiv author list rather than "and others." We also corrected the previously noted journal-entry type issue for `mcdermott2011sound`.

Change in manuscript: `reference.bib`: high-risk bibliographic entries corrected; remaining venue metadata should be checked again before final camera-ready submission.

m7: Figure 1 Caption is Vague

Reviewer Comment: The system diagram caption does not specify the DSP effect chain topology.

Our Response: Updated to explicitly list the nine-module topology: Compressor → Driver → Screamer → Equaliser → Chorus → Flanger → Phaser → Delay → Reverb, with each module switchable on/off.

Change in manuscript: Figure 1 caption rewritten.

m8: Missing Fusion Score Formula

Reviewer Comment: The fusion mechanism is described only verbally; no equation is provided.

Our Response: Added explicit formula: $s_{\text{fused}}(z_i) = \alpha s_{\text{text}}(z_i) + (1-\alpha)s_{\text{audio}}(z_i)$, where α is the text-side weight after score normalization. The manuscript now states that the adaptive profile is evaluated only as a robustness diagnostic.

Change in manuscript: Fusion formula added in the "Multimodal Fusion as a Supplementary Diagnostic Extension" subsection.

Summary of Changes

Issue	Type	Action Taken
M1 Dataset scale	Major	Near-dup sensitivity analysis + supplementary table
M2 Weak baselines	Major	PANNs, PaSST, MLP-Regressor added
M3 No ablations	Major	3 ablation experiments + subsection
M4 Unnormalized metrics	Major	Norm. L2 column + param_ranges.json
M5 Listening test issues	Major	Condensed and reframed as exploratory perceptual context
m1 Algorithm dim error	Minor	Algorithm 1 corrected to d=32
m2 Over-hedging	Minor	15+ instances reduced
m3 Unreferenced Sec 3.3	Minor	Section deleted
m4 Table 1 no data	Minor	Table removed
m5 Missing visualization	Minor	t-SNE + Gram heatmap figures added
m6 BibTeX errors	Minor	High-risk entries corrected
m7 Vague Fig 1 caption	Minor	Caption rewritten with topology
m8 Missing fusion formula	Minor	Formula added

We believe these revisions substantially strengthen the paper and address all reviewer concerns. We remain happy to provide further clarification or additional experiments if needed.